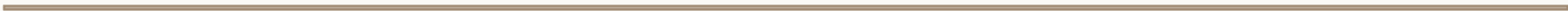


Would you still love me if I was a worm (who needed to learn Postgres)?

An amnesiac's guide to data modeling



This worm

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- Infrastructure and data platform engineer
- Distributed data, streaming, data warehouses - pg, kafka, delta lake, cassandra
- Former Data Infrastructure @ Patreon
- Starting new chapter ~cassandra~ @ Apple
- @seleniumk



You live underground.

**The dirt near the old root is better than the dirt
by the rock.**

Tommy knows what he did.



github.com/SeleniumK/worm-postgres

2

Getting Postgres running

just a couple commands



```
$ docker compose up -d    start Postgres
```

```
$ psql postgresql://worm:worm@localhost:5432/worm
```

```
worm=# \dt                list tables
```

```
worm=# \d worms           describe a table
```

```
worm=# \i file.sql        run a file
```

```
worm=# \q                quit
```

3

The problem, in sentences

before you type `CREATE TABLE`



I am a worm. I have met other worms.

I dig tunnels. Other worms dig tunnels.

I eat dirt, and I have opinions about it.

I am holding grudges. Some of them are very old.

The dirt near the old root is better than the dirt by the rock

Nouns are tables.

Verbs are relationships.

4

The 3am table

everything_i_remember



The table I made at 3am

```
CREATE TABLE everything_i_remember (  
  id                serial PRIMARY KEY,  
  who               varchar(50), -- names run long  
  tunnels_involved text,         -- 'Long Dark, Mudroom'  
  dirt_rating       real,        -- a float, to average  
  "timestamp"       timestamp,   -- no zone. quoted.  
  grudge            boolean,     -- can't hold a grudge  
  grudge_notes      text,        -- so, a second column  
  misc              text         -- the junk drawer  
);
```

```
worm=# \i 00-the-3am-table.sql
```

```
worm=# \d everything_i_remember
```

```
-- then 01-break-it.sql, one statement at a time
```

**Who am I currently holding
a grudge against, how bad is it,
when did it start,
and is it over?**

5

Types: the diagnosis

everything here just went wrong



Use this, not that

Use	Instead of	Because
text	varchar(n)	Austin
numeric	real, float8	4.1 was never 4.1
timestamptz	timestamp	no reference point
grudges table	boolean	a grudge has a direction
bigint identity	int, serial	more worms than you think
enum	free text	texture, severity
jsonb	misc text	the shape is unknown



**A vague type is not
a decision you avoided.
It is a decision you deferred,
with interest.**

6

Worms and tunnels

real entities, one-to-many



worms

```
CREATE TABLE worms (  
    id          bigint GENERATED ALWAYS  
                AS IDENTITY PRIMARY KEY,  
    name        text    NOT NULL,  
    length_mm   numeric(5,1),  
    is_me       boolean NOT NULL DEFAULT false,  
  
    CHECK (length_mm > 0)  
);
```

tunnels

```
CREATE TABLE tunnels (  
  id          bigint GENERATED ALWAYS  
              AS IDENTITY PRIMARY KEY,  
  name        text    NOT NULL UNIQUE,  
  dug_by_worm_id bigint NOT NULL  
              REFERENCES worms(id),  
  depth_cm    numeric(5,1),  
  dug_at      timestamptz  
);
```

```
worm=# \i 02-worms-and-tunnels.sql
```

```
worm=# \d tunnels
```

```
-- then 02a-break-the-fk.sql
```

**Constraints are documentation
the database actually enforces.**

7

Dirt tastings

a join table with its own facts



The textures of dirt

```
CREATE TYPE dirt_texture AS ENUM (  
    'loamy',  
    'gritty',  
    'suspiciously_damp',  
    'concerning'  
);
```

dirt_tastings

```
CREATE TABLE dirt_tastings (  
    id          bigint GENERATED ALWAYS  
                AS IDENTITY PRIMARY KEY,  
    worm_id     bigint NOT NULL REFERENCES worms(id),  
    tunnel_id   bigint NOT NULL REFERENCES tunnels(id),  
    tasted_at   timestamptz NOT NULL DEFAULT now(),  
    rating      numeric(3,1),  
    texture     dirt_texture,  
    notes       text,  
  
    CHECK (rating BETWEEN 0 AND 5)  
);
```

```
worm=# \i 03-dirt-tastings.sql
```

This is called a join table.

8

Grudges

a worm, pointing at a worm



Everyone was introduced by Stephen

```
ALTER TABLE worms
  ADD COLUMN introduced_by_worm_id bigint
  REFERENCES worms(id);
```

```
-- ...everyone gets Stephen. Then:
SELECT name FROM worms
WHERE introduced_by_worm_id IS NULL;
```

name

Stephen

grudges

```
CREATE TABLE grudges (  
    wronged_worm_id    bigint NOT NULL  
                        REFERENCES worms(id),  
    offending_worm_id  bigint NOT NULL  
                        REFERENCES worms(id),  
    severity            grudge_severity NOT NULL,  
    reason              text NOT NULL,  
    formed_at           timestamptz NOT NULL DEFAULT now(),  
    resolved_at         timestamptz,  
  
    PRIMARY KEY (wronged_worm_id, offending_worm_id),  
    CHECK (wronged_worm_id <> offending_worm_id)  
);
```

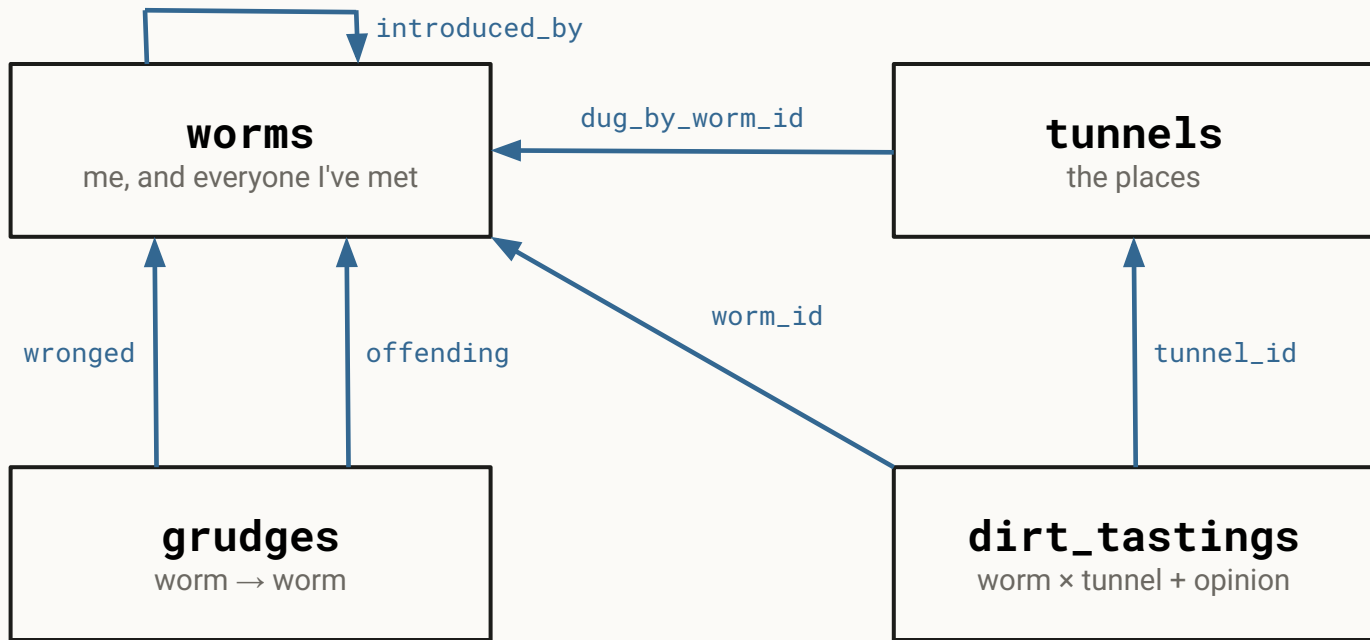
```
worm=# \i 04-grudges.sql
```

```
worm=# \d grudges
```

```
-- then 04a-break-the-constraints.sql
```

**That's a self-referencing
many-to-many.**

The worm's entire life



9

Asking the question you meant

four questions



Who do I owe an apology to?

Which tunnel has the best dirt?

**Which tunnels have I dug
but never eaten in?**

What do we owe each other?

10

Making it faster is someone else's problem

indexes, and what they cost



**An index is a second copy
of your data, sorted,
that Postgres maintains for you.**

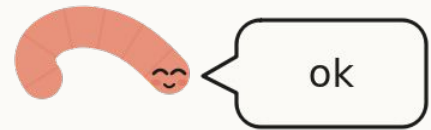
I am a worm. I have met other worms. → `worms`

I dig tunnels. Other worms dig tunnels. → `tunnels`

I eat dirt, and I have opinions about it. → `dirt_tastings`

I am holding grudges. Some are very old. → `grudges`

The dirt near the old root is better... → `rating, tunnel_id`



- Run the repo. Break something on purpose.
- Read the Postgres docs on data types.
- Next time you make a table, write four sentences first.



github.com/SeleniumK/worm-postgres

